

O'REILLY® Live Course Proposal

**Title: Hand-On Agentic GraphRAG**

Subtitle: Build Memory-Rich AI Agents with Graph-Based Retrieval, Temporal Memory, and Tool Orchestration

Acquisitions Editor to Complete:

Role: LLM Engineer, ML Engineer, Data Scientist, Applied AI Engineer, Knowledge Graph Engineer, Software Engineer

Level: Intermediate to Advanced

Skills: RAG, GraphRAG, LLMs, Knowledge Graphs, AI Agents

- Model an enterprise domain as a knowledge graph
- Build a simplified GraphRAG indexing pipeline
- Implement query-time GraphRAG retrieval operators
- Design graph-based memory
- Evaluate and harden an Agentic GraphRAG system

Contacts

Provide the contact information for the talent, and in the event of multiple presenters, duplicate the 'Instructor' and 'Entity' sections below and enter all information for each presenter separately.

Instructor Name: Ammar Mohanna
Email: mohannaammar@gmail.com
Phone: +961 3 916 092
Timezone: (GMT+2)
Mailing Address: Jnah, Beirut, Lebanon
Are you a returning or new instructor? Returning

Acquisitions Editor: Nicole Butterfield
Instructional Designer:

Timing and Scheduling

Indicate below the planned number of days per course, the duration of each day, and then the total duration (all days combined).

Number of day(s) per course: 1 (almost always 1)

Duration per day: 5 hours (max 4 hours)

Total Duration: 5 hours

Instructor's general availability in Pacific Time (mornings PT are optimal)

- Usually available, no constraints.

To be completed by Production and Editorial during scheduling:

Initial dates

- Aug 4, 2025

Presentation slides due date

- Jul 7, 2025
- [Recommended slide template](#)

Course Information

Supporting Technologies

ON24 Tools *ON24 is the platform that hosts our live online events. Please reference [this document](#) to see the full list of tools you may use during your course.*

- This live course will utilize O'Reilly's platform tools to engage learners, including live Q&A, polls, pulse checks, and discussion prompts throughout each session. The instructor may also use on-screen annotation during code walkthroughs for clarity.
- All code examples and exercise files will be shared via a GitHub repository (link to be provided). Learners can download the materials and run them locally to follow along. Resources such as slides or worksheets (if any) will be available for download as well.

Will this course use JupyterHub notebooks? See [this guide](#).

Yes / No No. (The instructor will use Google Colab instead. Notebooks and any required data/files will be provided through GitHub)

Course Registration Page Information

Note: The highlighted information below will be included in the course registration page for users to determine if the course is a good fit for their needs. It will be copyedited for grammar and clarity.

Course Outcomes This includes measurable skills that the learner will be able to use after completing the course.

By completing this live course, you will be able to:

- Explain why vector-only RAG fails on multi-hop, global, and time-sensitive enterprise questions, especially inside agent loops
- Design a lightweight knowledge graph schema for an enterprise domain (services, incidents, runbooks, owners, changes)
- Build a simplified GraphRAG index: entity graph plus precomputed community summaries for global questions
- Implement query-time GraphRAG operators: entity linking, constrained traversal, evidence-path context building, and hybrid vector + graph retrieval
- Wrap GraphRAG in an observe-think-act agent loop that chooses tools deliberately (not randomly) and updates memory after actions
- Implement graph-based memory primitives (working, episodic, semantic) with temporal awareness and confidence/provenance metadata
- Create a practical evaluation harness (golden questions + regression checks) to keep your Agentic GraphRAG system stable as data changes

Course Description

Agentic AI systems do not just answer questions, they pursue goals through observe-think-act loops and must remember what happened yesterday, understand how systems relate, and choose the right tools at the right time. In this hands-on course, we will build an “Autonomous Incident Commander” for a fictional microservice platform. You will start with a baseline vector RAG chatbot and watch it fail on the exact kinds of questions real operators ask: multi-hop dependency questions, global “what changed?” questions, and time-sensitive troubleshooting.

Then we will upgrade the architecture to Agentic GraphRAG. We will model the domain as a knowledge graph, build a GraphRAG index (including community summaries for global queries), implement graph-guided retrieval, and finally wrap it inside an agent loop with graph-based memory. Throughout the day you will produce a working notebook pipeline you can reuse on your own data and a checklist for production hardening.

Course Objectives What will the attendees learn and how could they apply it?

By the end of this live course, participants will be able to:

- Translate an enterprise knowledge domain into entities, relationships, and temporal events that support reasoning
- Choose an appropriate graph representation (property graph vs RDF vs hybrid) based on reasoning needs and latency constraints
- Build and query a GraphRAG pipeline that supports both local lookups and global synthesis
- Design an agent loop that uses GraphRAG outputs as structured context and logs decisions into memory
- Add temporal reasoning so the agent can prefer current truth over stale facts
- Evaluate retrieval and answer quality with a repeatable, automated harness to prevent regressions

Target Audience This section should detail the target audience. This can be a specific role (ie: software architects) or based on a learner's goal (ie: You want to be able to use X tool in your project).

This course is for practitioners building LLM systems in production who are hitting the limits of vector-only RAG:

- ML engineers and LLM engineers building RAG or agentic workflows
- Data scientists and applied researchers working on enterprise search, copilots, or automation
- Software engineers integrating LLMs with tool APIs, catalogs, and operational knowledge
- Knowledge graph practitioners who want a practical GraphRAG and agents workflow

The examples focus on enterprise reasoning and operations, but the patterns apply to any domain where relationships and time matter (security, compliance, finance, healthcare, product analytics).

Prerequisites What prior knowledge or experience is necessary?

To get the most out of this course, participants should have the following prior knowledge or experience:

- Comfortable Python skills (read code, run notebooks, basic data structures)
- Basic familiarity with LLMs and RAG concepts (embeddings, retrieval, prompting) is helpful
- No prior knowledge graph experience is required, but it helps

Note: Labs are designed to run in Colab without paid API keys. Optional cells show how to swap in a hosted LLM if you have credentials.

Course Preparation Any setup instructions? Any supplemental materials, like resources on O'Reilly Learning (including other Live Courses), PDF worksheets, or links to code repositories?

Recommended (best experience): use O'Reilly JupyterHub provided for the course, no local installs needed.

Optional local setup (if learners want to run everything on their machine):

- Python 3.10+
- Ability to run Jupyter notebooks locally
- Git installed to clone the repo (repo link TBD and shared before the course)

What learners will receive:

- A GitHub repo (link TBD) containing:
- 5 notebooks (one per session)
- Small datasets or download scripts
- “Fast mode” settings so training finishes live on CPU
- A short README with troubleshooting tips

Course Follow-Up Links to content in our platform for further learning. [Search here](#).

- Take [Knowledge Graphs & Large Language Models Bootcamp](#) (Live Event)
- Read [Agentic Graph RAG](#) (book)
- Read [GraphRAG: The Definitive Guide](#) (book)

Related Live Courses Editor: Please indicate pathing by noting any other courses — from this instructor or others — that could logically be taken before or after this course (if any).

- **Before** this course:
 - [Fundamentals of Large Language Models](#)
 - [Hands-on Retrieval Augmented Generation \(RAG\)](#)
- **After** this course:
 - [AI Agent Design Patterns](#)

Related Certification Note if this course helps prepare for a specific certification exam.

- N/A

Course schedule

The time frames are only estimates and may vary according to how the class is progressing.

Day 1: Hands-On Agentic GraphRAG (5 hours)

Baseline vector RAG and the five failure modes (50 minutes)

- Presentation: Course goals, repository and notebook orientation; incident story setup; why vector-only RAG breaks down inside agents (context amnesia, relationship blindness, temporal ignorance, reasoning paralysis, tool chaos); local vs global questions.

- Hands-on exercises: Run the starter notebook and confirm the dataset loads; execute three baseline vector RAG queries (local, multi-hop, global) and inspect retrieved chunks; create a small challenge set for regression testing; quick poll and chat share-out on observed failures.
- Q&A

Break (10 minutes)

Modeling the domain as a knowledge graph (50 minutes)

- Presentation: Choosing a graph model (labeled property graphs vs RDF) and when a hybrid makes sense; schema patterns for agentic systems (identity resolution, provenance, temporal change); designing for retrieval operators (traversal and explainability).
- Hands-on exercises: Build a graph from pre-extracted entities and relationships (NetworkX-based) and validate the schema; add temporal fields (valid_from/valid_to) for config changes and incidents; write traversal queries (two-hop and three-hop) that the baseline vector bot could not answer; pulse check: share one traversal path that answers a multi-hop question.
- Q&A

Break (10 minutes)

GraphRAG retrieval in practice (50 minutes)

- Presentation: Indexing intuition: entity graph plus community summaries for global synthesis; query-time operators (entity linking, constrained traversal, neighborhood expansion, hybrid vector and graph seeding); context synthesis for traceable answers (evidence paths and compact summaries).
- Hands-on exercises: Implement a simplified GraphRAG retriever (seed nodes, traverse, rank, assemble context); use community summaries to answer a global question (for example, "what changed across the platform this week?"); compare baseline vs GraphRAG on the challenge set and record improvements; mini-challenge: tune one operator (hop limit, edge types, ranking weights) and report the impact on one question.
- Q&A

Break (10 minutes)

Adding agency: observe-think-act plus graph-based memory (50 minutes)

- Presentation: From RAG to agentic workflows: why tool and skill routing must be explicit and auditable; skills integration and protocol standardization covering MCP-style skill manifests (capability discovery, schemas, permissions) and A2A delegation patterns for composing specialist skills; memory types (working, episodic, semantic, procedural) and what each stores; graph memory patterns (temporal awareness, confidence and provenance metadata, hot-warm-cold lifecycle).

- Hands-on exercises: Wrap GraphRAG in a simple agent loop that routes between skills (vector lookup, graph traversal, runbook lookup) via a lightweight skill registry plus MCP-style manifests; optionally delegate one subtask to a specialist agent (A2A style) and merge results; log actions, decisions, and outcomes into an episodic memory graph and use it to answer “what did we try last time?”; add a temporal filter so the agent prefers current configuration facts over stale ones; chat share-out: one memory edge (action to outcome) and how it changed a follow-up response.
- Q&A

Break (10 minutes)

Hardening for production (50 minutes)

- Presentation: Evaluation: golden questions, a regression harness, and how to detect retrieval drift as the graph evolves; operational concerns (latency budgets, caching, incremental updates, schema evolution); governance basics (access control, provenance, audit logs, safe tool authority boundaries).
- Hands-on exercises: Build an evaluation harness that runs baseline vs GraphRAG vs agent answers on the challenge set; introduce a simulated new incident update and verify the system adapts without regressions; write a personal production checklist (what to log, what to monitor, what to test); final reflection in chat: one change you will make to your RAG or agent system next week.
- Q&A

Wrap-up and Q&A (10 minutes)

Instructor Information

Bio

Ammar Mohanna is a seasoned AI expert, educator, and entrepreneur with extensive experience spanning academia, industry consulting, and technology innovation. He holds a PhD in Edge Artificial Intelligence from the University of Genoa, Italy. Ammar currently serves as a Lecturer at the American University of Lebanon, where he teaches advanced courses in AI and Machine Learning, helping shape the next generation of AI professionals.

As an AI Consultant, Ammar leads initiatives focused on integrating AI and Generative AI into educational technologies, collaborating closely with cloud technologies. His professional expertise includes Machine Learning, MLOps, Explainable AI (XAI), Kubernetes, and Microservices architecture. Previously, Ammar was the AI Lead at Assentify, a company he co-founded dedicated to providing specialized AI solutions, training, and consultancy.

Ammar actively contributes to the AI community through workshops, mentorship, and public speaking engagements, focusing on the practical adoption and strategic implementation of AI across various sectors. He has also developed numerous online courses and training modules, emphasizing hands-on and accessible AI education.

Ammar is fluent in Arabic, English, and French, with intermediate proficiency in Italian, and currently resides in Beirut, Lebanon.

Company affiliation

AI Consultant and Lecturer

Social media URLs

Primary website	https://ammarmohanna.ai/
Twitter	https://x.com/AmmarYMohanna
LinkedIn	https://www.linkedin.com/in/ammarmohanna/
GitHub	https://github.com/AmmarMohanna
YouTube	
Other	

Video samples of you engaged in teaching

- [Testing AI Systems: Crafting Trust, Quality, and Performance](#)

Testimonials such as praise from previous clients or glowing tweets

- <https://ammarmohanna.ai/>

Instructor photo

- See [our guide to taking a good headshot](#).
- We love to have a few options to select from; send us as many as you can.
- Format: JPG or PNG, ideally high-res, near 1400x1400 pixels (original files are best)
- Please add a link to your headshot(s) here (Dropbox, Google Drive, etc.) or email files directly to your editor or event producer.
- **Do not paste photos in this Google Doc.** (We can't use them!)
- Acquisition Editors: Rename headshot as *Firstname Lastname*, then copy into [Headshot Inbox](#).

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Proposal Checklist

Minimum requirements checklists. All O'Reilly courses must include the following.
See the [Guide for Instructors](#) for context.

Course info

- ☐ Instructor contact info captured
- ☐ Initial timing and duration values captured
- ☐ All course info fields filled in
- ☐ Confirmed with instructor intent to use interactive labs or challenges
- ☐ Tested and verified links to all platform-based content and other materials

Instructor info

- ☐ Bio and social media links captured
- ☐ New instructors only:
 - ☐ Video samples
 - ☐ Testimonials
 - ☐ Headshot submitted

Learning plan

- ☐ Description communicates the value of this learning (the "why")
- ☐ Learning promise elements are clearly articulated
- ☐ Pework well defined (if any) and all needed materials linked to
- ☐ Resources for further learning listed and linked to
- ☐ Schedule alternates presentations and activities; no lectures
- ☐ Schedule accounts for short breaks every hour
- ☐ Schedule indicates durations for each section, and durations total sum as expected
- ☐ Incorporated [engagement best practices](#)
- ☐ Each segment has time for Q&A
- ☐ Exercises provide opportunities for hands-on problem solving
- ☐ All exercises include an assessment component (via chat)
- ☐ Any external services used comply with our privacy policy